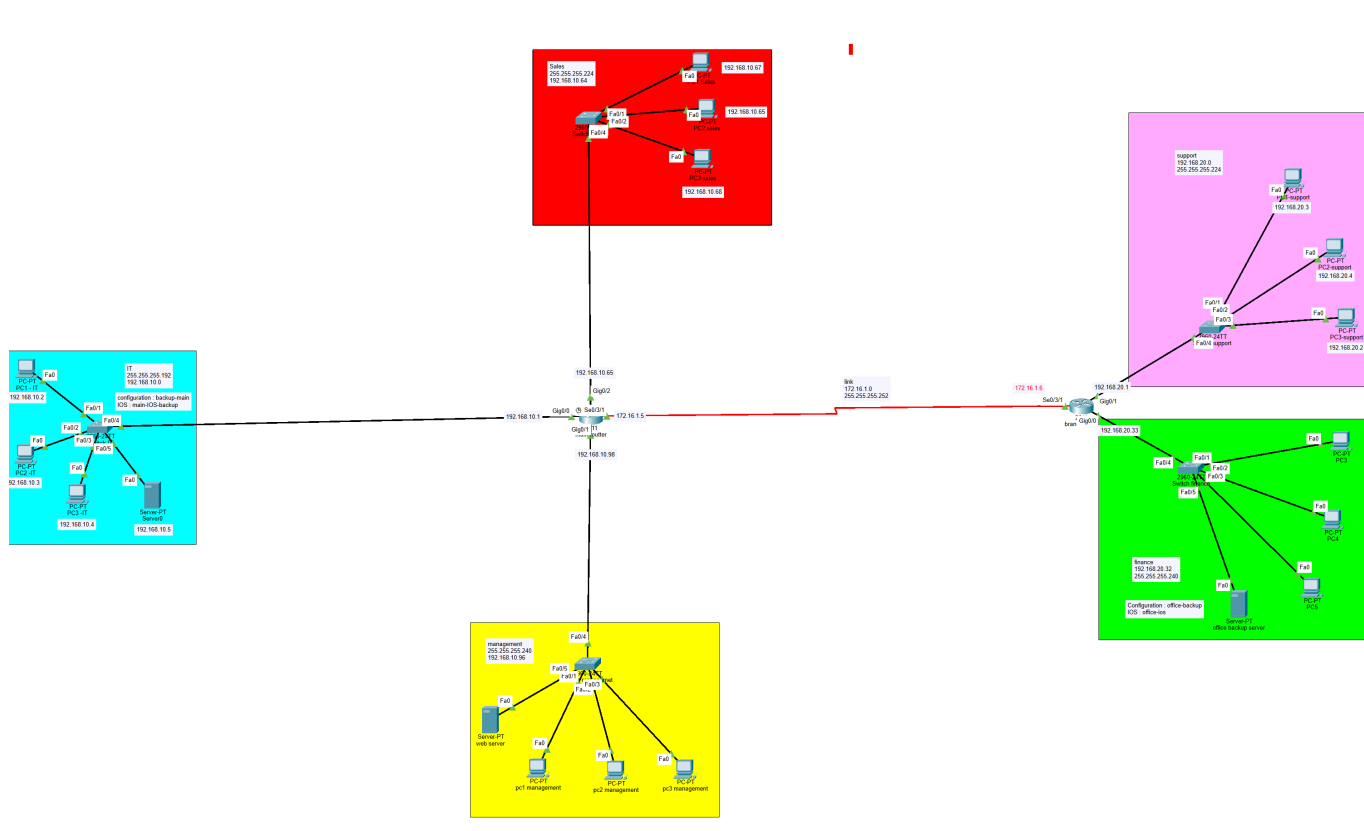


# Packet tracer project

project Name : Full CCNA 200-301 Finish project

due date : from 1 July 2026 to 20 July 2026

by : Ahmed Mostafa ( cyber security analyst)



## 1. Subnets ( main network)

| Subnet Name | Subnet mask     | Network IP    | Broadcast IP   | Number of hosts |
|-------------|-----------------|---------------|----------------|-----------------|
| IT          | 255.255.255.192 | 192.168.10.0  | 192.168.10.63  | 62              |
| Sales       | 255.255.255.224 | 192.168.10.64 | 192.168.10.95  | 30              |
| Management  | 255.255.255.240 | 192.168.10.96 | 192.168.10.111 | 14              |

## 2. Subnets ( branch network)

| Subnet Name | Subnet mask     | Network IP    | Broadcast IP  | Number of hosts |
|-------------|-----------------|---------------|---------------|-----------------|
| Support     | 255.255.255.224 | 192.168.20.0  | 192.168.20.31 | 30              |
| Finance     | 255.255.255.240 | 192.168.20.32 | 192.168.20.47 | 14              |

### 3. Routers

| Name          | Se0/3/1               | Gig0/0                        | Gig0/1                            | Gig0/2                       |
|---------------|-----------------------|-------------------------------|-----------------------------------|------------------------------|
| Main Router   | 172.16.1.5<br>(WAN 1) | 192.168.10.1 (IT-LAN)         | 192.168.10.98<br>(Management-LAN) | 192.168.10.65<br>(Sales-LAN) |
| Branch Router | 172.16.1.6<br>(WAN 1) | 192.168.20.1<br>(Support-LAN) | 192.168.20.33<br>(Finance-LAN )   | -----                        |

#### Steps :

1. implement a Design for topology and specify branches
2. Divide each network to branches ( subnets ) using VLSM to Save ip addresses
3. change routers hostnames

Router 1 :hostname main-Router

Router 2: hostname branch-Router

4. set interfaces IP address and open interfaces with the command `no shutdown`
5. Enable DHCP on routers to get IP addresses dynamically
6. enable OSPF Routing for each router

main-router

`O 192.168.20.0/27 [110/65] via 172.16.1.6, 00:17:11, Serial0/3/1`

`O 192.168.20.32/28 [110/65] via 172.16.1.6, 00:17:11, Serial0/3/1`

branch router `O 192.168.10.0/26 [110/65] via 172.16.1.5, 00:19:23, Serial0/3/1` `O 192.168.10.64/27 [110/65] via 172.16.1.5, 00:19:23, Serial0/3/1` `O 192.168.10.96/28 [110/65] via 172.16.1.5, 00:19:23, Serial0/3/1``

7. enable Extended access control list block pc 3 on finance subnet from access http on web server in management subnet on interface Gig0/0 on branch router

`access-list 101 deny tcp host 192.168.20.36 host 192.168.10.101 eq 80`

`access-list 101 IP permit any`

`int Gig0/0`

`IP access group 101 in`

8. set main router as NTP server for branch router

configuration on main router :

`set clock <Time><Date>`

`NTP master`

configuration on branch router

```
ntp server 172.16.1.5
```

9. Set backup server for main network and backup server for branch

main router server IP : 192.168.10.5

```
configuration file backup
```

```
Copy run tftp
```

```
192.168.10.5
```

```
backup-main
```

```
IOS file backup
```

```
copy flash tftp
```

```
192.168.1.5
```

```
IOS file name .bin
```

```
main-IOS-backup
```

main router server IP : 192.168.20.40

```
configuration file backup
```

```
Copy run tftp
```

```
192.168.20.40 backup-main IOS file backup copy flash tftp 192.168.20.40 IOS file  
name .bin office-IOS
```

## Verification and Testing

1. Success Ping between all subnets
2. Trace Route
3. make sure all the interface is UP/UP using command  

```
show ip interface brief
```

### main-router

#### Interface IP-Address OK? Method Status Protocol

```
GigabitEthernet0/0 192.168.10.1 YES NVRAM up up  
GigabitEthernet0/1 192.168.10.98 YES NVRAM up up  
GigabitEthernet0/2 192.168.10.65 YES NVRAM up up  
Serial0/3/1 172.16.1.5 YES manual up up
```

### branch-router

#### Interface IP-Address OK? Method Status Protocol

```
GigabitEthernet0/0 192.168.20.33 YES manual up up  
GigabitEthernet0/1 192.168.20.1 YES manual up up  
Serial0/3/1 172.16.1.6 YES manual up up
```